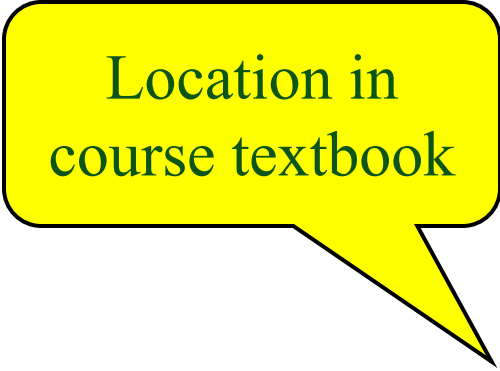


NUMBER SYSTEMS & DATA REPRESENTATION

Week # 07

A yellow speech bubble with a black outline and a tail pointing towards the bottom right. It contains the text "Location in course textbook".

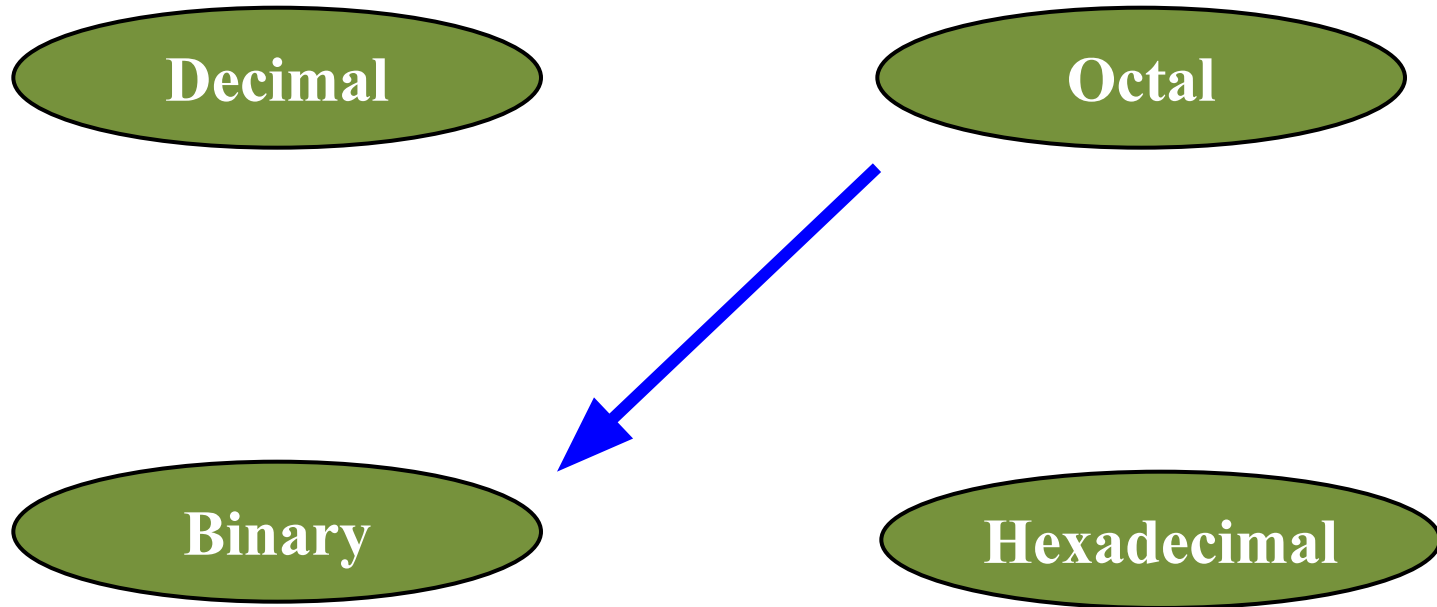
Location in
course textbook

Chapt. 3

CONVERSION OF OCTAL, HEXADECIMAL TO BINARY

- The conversion of a number from octal and hexadecimal to binary uses the inverse of the steps defined for the conversion of binary to octal and hexadecimal.

Octal to Binary

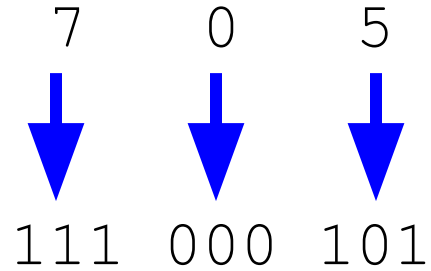


Octal to Binary

- **Algorithm**
 - **Convert** each octal digit to a **3-bit** equivalent binary representation.

Example

$$705_8 = ?_2$$



$$705_8 = 111000101_2$$

Hexadecimal to Binary

Decimal

Octal

Binary

Hexadecimal



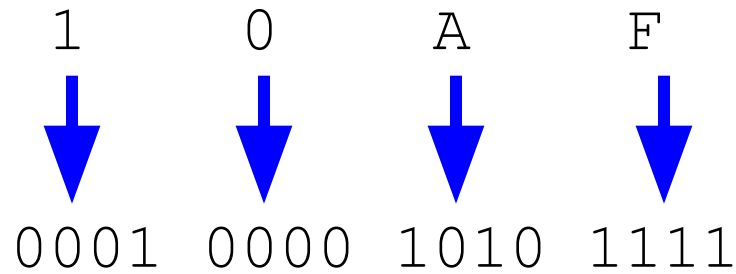
Hexadecimal to Binary

- **Algorithm**

- **Convert** each hexadecimal digit to a **4-bit** equivalent binary representation.

Example

$$10AF_{16} = ?_2$$



$$10AF_{16} = 0001000010101111_2$$

Class Task

- Convert the octal number 473 to **binary**.
- Convert the hexadecimal number 2BA3 to **binary**.

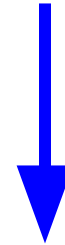
Octal to Hexadecimal

Decimal

Octal

Binary

Hexadecimal



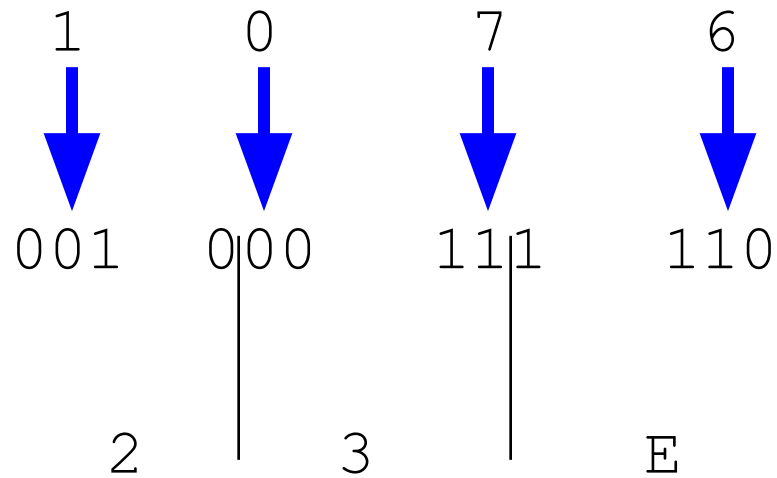
Octal to Hexadecimal

- **Algorithm**

- Use **binary** as an **intermediary**.
- **Convert** binary into hexadecimal.

Example

$$1076_8 = ?_{16}$$



$$1076_8 = 23E_{16}$$

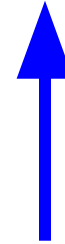
Hexadecimal to Octal

Decimal

Octal

Binary

Hexadecimal



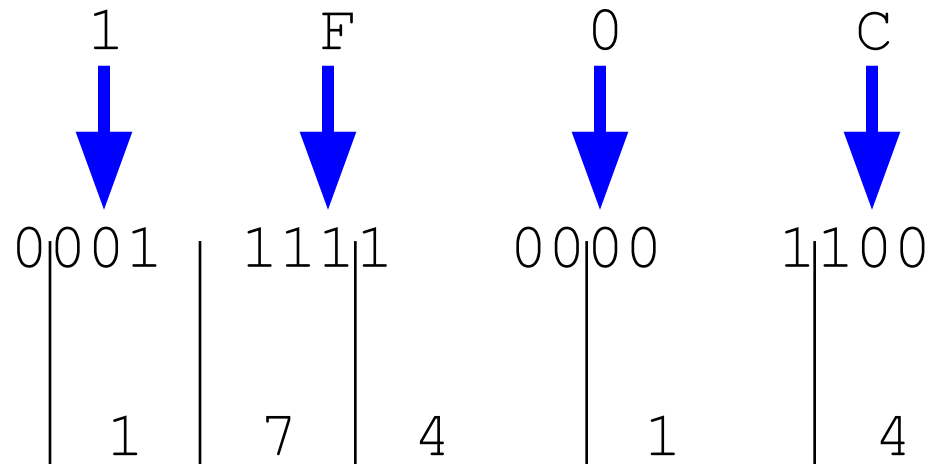
Hexadecimal to Octal

- **Algorithm**

- Use binary as an **intermediary**.
- Group bits in **threes**, starting on **right**.
- **Convert** binary into decimal.

Example

$$1F0C_{16} = ?_8$$



$$1F0C_{16} = 17414_8$$

Exercise – Convert ...

Decimal	Binary	Octal	Hexa- decimal
33			
	1110101		
		703	
			1AF

Don't use a calculator!

Skip answer

Answer

Exercise – Convert ...

Answer

Decimal	Binary	Octal	Hexa- decimal
33	100001	41	21
117	1110101	165	75
451	111000011	703	1C3
431	110101111	657	1AF

EXERICSE – CONVERT...

Decimal	Binary	Octal	Hexa- decimal
29.8			
	101.1101		
		3.07	
			C.82

Don't use a calculator!

Skip answer

Answer

Exercise – Convert ...

Answer

Decimal	Binary	Octal	Hexa- decimal
29.8	11101.110011...	35.63...	1D.CC...
5.8125	101.1101	5.64	5.D
3.109375	11.000111	3.07	3.1C
12.5078125	1100.10000010	14.404	C.82

Common Powers (1 of 2)

- **Base 10**

Power	Preface	Symbol	Value
10^{-12}	pico	p	.0000000000001
10^{-9}	nano	n	.000000001
10^{-6}	micro	μ	.000001
10^{-3}	milli	m	.001
10^3	kilo	k	1000
10^6	mega	M	1000000
10^9	giga	G	1000000000
10^{12}	tera	T	1000000000000

Common Powers (2 of 2)

- **Base 2**

Power	Preface	Symbol	Value
2^{10}	kilo	k	1024
2^{20}	mega	M	1048576
2^{30}	Giga	G	1073741824

- In computing, particularly w.r.t. memory, the **base-2** interpretation generally applies.

Review – multiplying powers

- For common bases, **add powers**.

$$a^b \times a^c = a^{b+c}$$

$$2^6 \times 2^{10} = 2^{16} = 65,536$$

or...

$$2^6 \times 2^{10} = 64 \times 2^{10} = 64k$$

Binary Addition (1 of 2)

- Two 1-bit values

A	B	A + B
0	0	0
0	1	1
1	0	1
1	1	10

“two”

Input 1	Input 2		Sum	Carry
0	0	→	0	No carry
0	1	→	1	No carry
1	0	→	1	No carry
1	1	→	0	1

Table 5.3 Binary addition rules

Input 1	Input 2	Input 3		Sum	Carry
0	0	0	→	0	No Carry
0	0	1	→	1	No Carry
0	1	0	→	1	No Carry
0	1	1	→	0	1
1	0	0	→	1	No Carry
1	0	1	→	0	1
1	1	0	→	0	1
1	1	1	→	1	1

Table 5.4 Binary addition of three inputs

Binary Addition (2 of 2)

- Two *n*-bit values
 - Add individual bits
 - Propagate carries
 - E.g.,

$$\begin{array}{r} 10101 \\ + 11001 \\ \hline 101110 \end{array}$$

$$\begin{array}{r} 21 \\ + 25 \\ \hline 46 \end{array}$$

EXAMPLE

- Add 01011101_2 and 00110010_2
- **Solution:**

$$\begin{array}{rcccccccc} & & & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & \\ + & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & \\ \hline & & & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{array}$$

EXAMPLE

- Add **111101** and **10111**.

	1	1	1	1	1	1		
		1	1	1	1	0	1	= 61
+			1	0	1	1	1	= 23
	1	0	1	0	1	0	0	= 84

Class tasks (Binary Addition)

1. Add 10 and 01. Verify the answer with the help of decimal addition.
2. Add 01 and 11. Verify the answer with the help of decimal addition.
3. Add 11 and 11. Verify the answer with the help of decimal addition.
4. Add 1101 and 1111. Verify the answer with the help of decimal addition.
5. Add 10111, 11100 and 11. Verify the answer with the help of decimal addition.

Solutions

Binary Addition	Decimal Addition
$\begin{array}{r} 10 \\ + 01 \\ \hline \text{Result } 11 \end{array}$	$\begin{array}{r} 2 \\ + 1 \\ \hline 3 \end{array}$
$11_2 = 3_{10}$	

Binary Addition	Decimal Addition
$\begin{array}{r} 11 \leftarrow \text{Carry} \\ 01 \\ + 11 \\ \hline \text{Result } 100 \end{array}$	$\begin{array}{r} 1 \\ + 3 \\ \hline 4 \end{array}$
$100_2 = 4_{10}$	

Binary Addition	Decimal Addition
$\begin{array}{r} 11 \leftarrow \text{Carry} \\ 11 \\ + 11 \\ \hline \text{Result } 110 \end{array}$	$\begin{array}{r} 3 \\ + 3 \\ \hline 6 \end{array}$
$110_2 = 6_{10}$	

Binary Addition	Decimal Addition
$\begin{array}{r} 1111 \leftarrow \text{Carry} \\ 1001 \\ + 1111 \\ \hline 11000 \end{array}$	$\begin{array}{r} 9 \\ + 15 \\ \hline 24 \end{array}$
$11000_2 = 24_{10}$	

Binary Addition	Decimal Addition
$\begin{array}{r} 11111 \leftarrow \text{Carry} \\ 10111 \\ + 11000 \\ \hline 110110 \end{array}$	$\begin{array}{r} 23 \\ + 24 \\ \hline 47 \end{array}$
$110110_2 = 54_{10}$	

Binary Subtraction

Input 1	Input 2		Difference	Borrow
0	0	→	0	No borrow
0	1	→	1	1
1	0	→	1	No borrow
1	1	→	0	No borrow

Table 5.5 Binary subtraction rules

Binary Subtraction

Example 1: Subtract 01 from 11. Verify the answer with the help of decimal subtraction.

Binary Subtraction	Decimal Subtraction
$\begin{array}{r} 11 \\ - 01 \\ \hline \text{Result } 10 \end{array}$	$\begin{array}{r} 3 \\ - 1 \\ \hline 2 \end{array}$
$10_2 = 2_{10}$	

Binary Subtraction

Example 2: Subtract 01 from 10. Verify the answer with the help of decimal subtraction.

Binary Subtraction	Decimal Subtraction
$\begin{array}{r} 010 \\ + 0 \\ - 01 \\ \hline 01 \end{array}$	$\begin{array}{r} 2 \\ - 1 \\ \hline 1 \end{array}$
$01_2 = 1_{10}$	

Example 3: Subtract 0111 from 1110. Verify the answer with the help of decimal subtraction

Binary Subtraction	Decimal Subtraction
$ \begin{array}{r} 010 \\ 010 \\ 010 \\ \hline 1110 \\ - 0111 \\ \hline 0111 \end{array} $	$ \begin{array}{r} 14 \\ - 07 \\ \hline 7 \end{array} $
$0111_2 = 7_{10}$	

Example 4: Subtract 100110 from 110001. Verify the answer with the help of decimal subtraction.

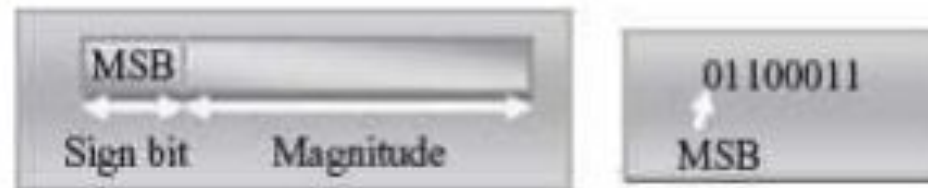
Binary Subtraction	Decimal Subtraction
$ \begin{array}{r} 11 \\ 1010 \\ 1010 \\ 1010 \\ \hline 110001 \\ - 100110 \\ \hline 001011 \end{array} $	$ \begin{array}{r} 49 \\ - 38 \\ \hline 11 \end{array} $
$001011_2 = 11_{10}$	

REPRESENTATION OF NEGATIVE NUMBERS

- There are many methods of representing **signed** number in **binary**:
 - **Sign-Magnitude representation Method**
 - **1's Complement representation Method**
 - **2's Complement representation Method**

1- Sign-Magnitude representation Method

- The sign of a binary number has to be represented using 0 and 1, in the computer.
- An n-bit signed binary number consists of two parts—sign bit and magnitude.
- The left most bit, also called the Most Significant Bit (MSB) is the sign bit. The remaining n-1 bits denote the magnitude of the number.
- For example, 01100011 is a positive number since its sign bit is 0, and, 11001011 is a negative number since its sign bit is 1



2- 1's COMPLEMENT METHOD

- **1's complement** of a **binary** number can be directly obtained by **changing** all **0's** to **1's** and all **1's** to **0's**.

- **Example:**

Take 1's complement of the binary number 01100110 directly.

Original number **01100110**

1's Complement **10011001**

3- 2's COMPLEMENT METHOD

- Algorithm

- First taking 1's complement.
- And then adding 1 in the result.

Example of 2's complement

- Take **2's complement** of the binary number **01100110** directly.

Original number **01100110**

1's Complement **10011001**

+1

2's Complement **10011010**

Example with all methods: Representation of the signed number 18

+18	0 0010010		Sign bit is 0. 0010010 is binary equivalent of +18
	Signed magnitude representation	1 0010010	Sign bit is 1. 0010010 is binary equivalent of +18
-18	Signed 1's complement representation	1 1101101	Sign bit is 1. 1101101 is 1's complement of +18
	Signed 2's complement representation	1 1101110	Sign bit is 1. 1101110 is 2's complement of +18

Example

- Q 1: Represent -54_{10} in **1's complement form** using 8 bits.
- Q 2: Represent -54_{10} in **2's complement form** using 8 bits. **Modulus of $-54_{10} = 54$**
- Q 3: Represent -54_{10} in **sign-magnitude method** using 8 bits ??

Solution

2		54	
2		27	0
2		13	1
2		6	1
2		3	0
2		1	1
		0	1

$$54_{10} = 0110110_2$$

$$54_{10} = 00110110_2 \text{ (in 8 bits)}$$

So -54_{10} in 1's complement form:

$$-54_{10} = 11001001_2$$

Solution

$$\begin{array}{r|l} 2 & 54 \\ \hline 2 & 27 \\ \hline 2 & 13 \\ \hline 2 & 6 \\ \hline 2 & 3 \\ \hline 2 & 1 \\ \hline & 0 \end{array} \quad \begin{array}{l} 0 \\ 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{array}$$

$$54_{10} = 0110110_2$$

$$54_{10} = 00110110_2 \text{ (in 8 bits)}$$

Take 1's complement of

$$00110110$$

$$= 11001001$$

$$+ \underline{1}$$

$$= 11001010_2$$

Addition of Signed BINARY Numbers

- **Modern day** computers use **2's complement** or **1's complement** method for performing signed additions.
- **Example:**
Calculate **38 – 29** using 8-bit **1's complement** method.

$$38 + (-29)$$

i. Write both the numbers in binary form using 8 bits.

$$38_{10} = 00100110_2$$

$$29_{10} = 00011101_2$$

ii. Represent negative numbers in 1's complement

$$-29_{10} = 11100010_2$$

iii-Add the number & its 1's complement

$$\begin{array}{r} 00100110_2 \\ + 11100010_2 \\ \hline \text{Carry: } 1 _2 \\ \text{Add End Carry: } + 1 \\ \hline 00001001_2 \end{array}$$

EXAMPLE

- Calculate $-54 - 30$ using 8-bit 1's complement method.
- **Solution:**

$$(-54) + (-30)$$

$$54_{10} = 00110110_2$$

$$30_{10} = 00011110_2$$

Represent negative numbers in 1's complement

$$-54_{10} = 11001001_2$$

$$-30_{10} = 11100001_2$$

Add numbers in its 1's complement form

$$\begin{array}{r} 11001001_2 \\ + \underline{11100001_2} \\ \hline \text{Carry: } 1 10101010_2 \\ \text{Add End Carry:} + \underline{1} \\ \hline 10101011_2 \end{array}$$

Convert the 1's complement result into decimal. As MSB is 1 so it's a negative number

$$10101011_2 = 01010100 = -84$$

Example: Using 2's complement method

- Example 1: Add +5 and +10.
- Example 2: Add -5 and +10.
- Example 3: Add +5 and -10.
- Example 4: Add -5 and -10.